SmartBank Customer Churn Analysis

Phase 1: Data Gathering & Exploratory Data Analysis Report

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1. Introduction

This report shows the first phase of customer retention enhancement through an analytics project and the objective of this phase is to identify and gather relevant customer data through EDA and clean and preprocess the data for ML model development in phase 2.

The bank reported a decline in customer retention among young and business owners. The main task of this project is to identify at risk customers and inform targeted retention strategies.

2. Data Overview

Provided data contains 1000 customer records with 5 variables. The table below summarise the dataset.

Column	Data Type	Description	Sample Values
CustomerID	Integer	Unique customer identifier	1 – 1000
Age	Integer	Customer age in years	18 – 69
Gender	Categorical	Customer gender (M/F)	M, F
MaritalStatus	Categorical	Marital status	Single, Married, Divorced, Widowed
IncomeLevel	Categorical	Income bracket	Low, Medium, High

3. Exploratory Data Analysis

3.1 Descriptive Statistics

The following descriptive statistics summarise the key numerical variable in the dataset:

Variable	Mean	Median	Std Dev	Min	Max
Age	43.27	43.00	15.24	18	69

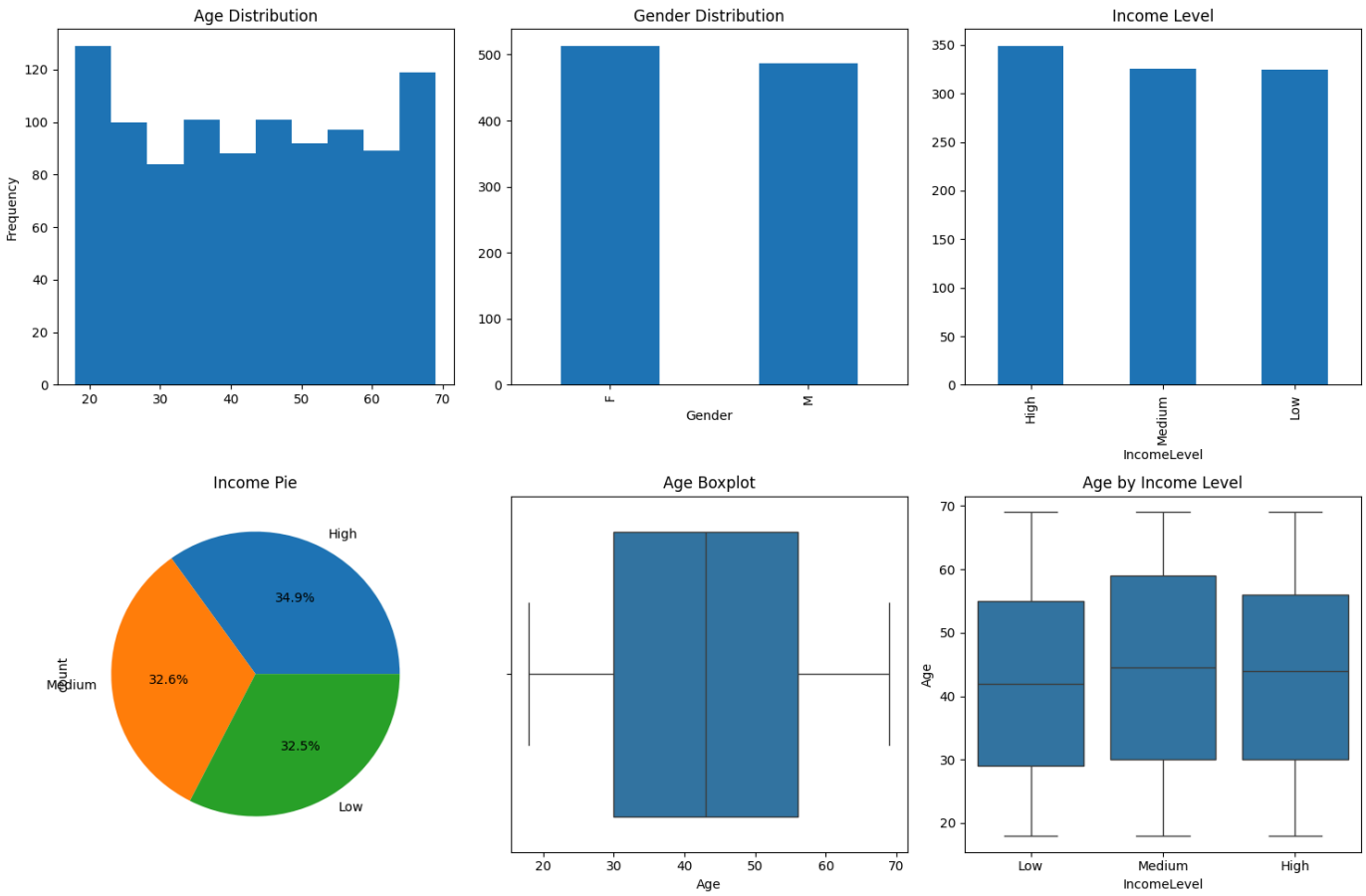
The average customer age is 43.27 years, with a standard deviation of 15.24, indicating a fairly wide spread of ages. The minimum age of 18 and maximum of 69 confirm the dataset spans both young professionals and older customers.

3.2 Categorical Variable Distributions

Variable	Category	Count	Percentage
Gender	Female (F)	513	51.3%
	Male (M)	487	48.7%
MaritalStatus	Widowed	276	27.6%
	Married	261	26.1%
	Divorced	248	24.8%
	Single	215	21.5%
IncomeLevel	High	349	34.9%
	Medium	326	32.6%
	Low	325	32.5%

3.3 Data Visualisations The following visualisations were generated to explore the distributions and relationships within the dataset:

Figure 1: EDA Visualisations — Age Distribution, Gender, Marital Status, Income Level, and Age by Group



3.4 Key Findings from EDA

- Age distribution is fairly uniform across the 18–69 range, with a mean of 43.3 years. The spread suggests the customer base spans multiple life stages, each with different churn risk profiles.
- Gender split is nearly equal (51.3% Female, 48.7% Male), indicating no significant gender imbalance in the dataset.
- Marital status is distributed across all four categories, with Widowed customers being the largest group (27.6%), followed closely by Married (26.1%), Divorced (24.8%), and Single (21.5%). This distribution is noteworthy as widowed and divorced customers may have experienced life events that affect banking behaviour.
- Income level is broadly balanced across Low (32.5%), Medium (32.6%), and High (34.9%) brackets, suggesting a diverse customer base with varying financial sensitivities.
- Box plot analysis shows that age varies across income levels and marital status categories, with no extreme outliers detected.

4. Data Cleaning and Preprocessing

4.1 Handling Missing Values

No missing values were identified in the dataset. All 1,000 records are complete across all five variables. Therefore, no imputation or deletion was required.

4.2 Outlier Detection and Treatment

Box plot analysis of the Age variable confirmed no extreme outliers. All ages fall within the plausible range of 18–69 years. No outlier treatment was necessary.

4.3 Encoding Categorical Variables

Machine learning algorithms require numerical inputs. The following encoding strategies were applied:

Variable	Method	Result
Gender	Binary Encoding	M=1, F=0 → new column: Gender_Encoded
MaritalStatus	One-Hot Encoding	4 binary columns: MS_Single, MS_Married, MS_Divorced, MS_Widowed
IncomeLevel	Ordinal Encoding	Low=0, Medium=1, High=2 → IncomeLevel_Encoded (ordinal relationship justified)

4.4 Standardisation of Numerical Features

The Age variable was standardised using Z-score standardisation (mean = 0, standard deviation = 1). This ensures that Age contributes proportionally alongside encoded variables during model training, preventing scale-based bias in distance-sensitive algorithms.

- Original Age mean: 43.27, Standard Deviation: 15.24
- Standardised column added: Age_Standardised

4.5 Final Cleaned Dataset Structure

After preprocessing, the cleaned dataset contains 1,000 rows and 12 columns:

Column	Type	Notes
CustomerID	Integer	Retained as identifier
Age	Integer	Original values retained
Gender	String	Original retained
MaritalStatus	String	Original retained
IncomeLevel	String	Original retained
Gender_Encoded	Integer (0/1)	Binary encoded Gender
MS_Divorced	Integer (0/1)	One-hot: MaritalStatus
MS_Married	Integer (0/1)	One-hot: MaritalStatus
MS_Single	Integer (0/1)	One-hot: MaritalStatus
MS_Widowed	Integer (0/1)	One-hot: MaritalStatus
IncomeLevel_Encoded	Integer (0–2)	Ordinal encoded IncomeLevel
Age_Standardised	Float	Z-score standardised Age

5. Observations and Hypotheses

- Hypothesis 1- Age and Churn: Younger customers (ages 18–30) may show higher churn rates, as this segment is more likely to explore competitor offerings and digital-first banking alternatives.

- Hypothesis 2 - Marital Status and Churn: Single and Divorced customers may be more prone to churn due to fewer financial commitments tying them to a single bank.
- Hypothesis 3 - Income Level and Churn: Low-income customers may be more sensitive to competitor pricing and personalised service offerings, making them higher churn risks.
- Hypothesis 4 - Gender: The near-equal gender split suggests gender alone is unlikely to be a dominant predictor, but may interact with other variables.

These hypotheses will be tested and validated during the model-building phase through feature importance analysis.

6. Limitations

- The dataset does not include a Churn label (target variable). Phase 2 will require a labelled dataset or a proxy variable (e.g., account closure, inactivity) to train a supervised classification model.
- Only demographic variables are present. Transaction history, customer service interactions, and product usage data - which are strong churn predictors - are absent and should be incorporated if available.
- The dataset size of 1,000 records is relatively small for training a robust machine learning model. A larger dataset would improve model reliability.

7. Conclusion

This report has documented the first phase of the SmartBank Customer Retention Enhancement project. The dataset was successfully gathered, explored, and preprocessed. Key demographic patterns were identified across age, gender, marital status, and income level. The data has been cleaned and prepared - with encoded categorical variables and standardised numerical features - and is now ready for machine learning model development in Phase 2.

The cleaned and preprocessed dataset has been submitted alongside this report as: Cleaned_Customer_Churn_Data.xlsx